

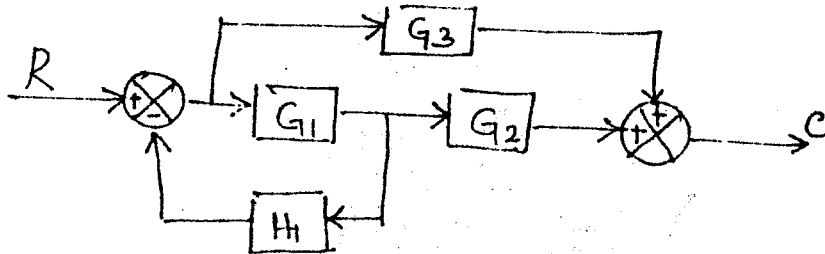
B. Tech Degree VI Semester (Supplementary) Examination June 2006

ME 605 CONTROL SYSTEMS (Prior to 1998 Admissions)

Time : 3 Hours

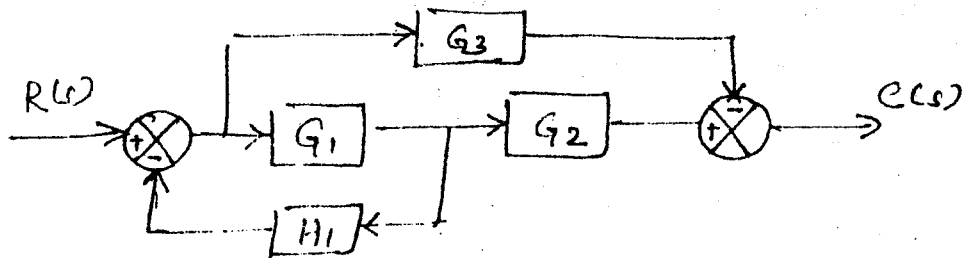
Maximum Marks : 100

- I. (a) (i) Define transfer function.
(ii) Differentiate between open loop and closed loop control systems. (10)
- (b) Reduce the block diagram and find $\frac{C}{R}$. (10)



OR

- II. (a) Convert the block diagram in to signal flow graph and determine $\frac{C(s)}{R(s)}$. (10)



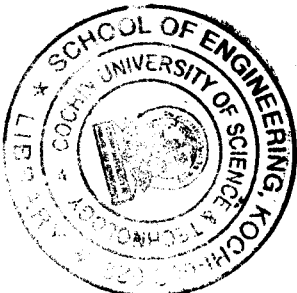
- (b) For the transfer function

$$G(s) = \frac{1}{2} \frac{(s^2 + 4)(1 + 2.5s)}{(s^2 + 2)(1 + 0.5s)}$$

Plot poles and zeros in s plane and determine the value of transfer function at $s = 0.2$. (10)

- III. (a) What is the importance of test signals? Define standard test signals. (10)

(Turn Over)



- (b) Laplace transform of error signal

$$E(s) = R(s) \frac{s}{s^2 + 6s + 25}$$

Calculate the steady state error for $r(t) = t$.

(10)

OR

- IV. (a) Sketch the response of a second order undamped system and mark the time domain specifications. Define these specifications. (10)

- (b) A unity feed back system has an open loop transfer function $G(s)H(s) = \frac{10}{s(s+2)}$.

Determine the rise time and percentage overshoot.

(10)

- V. (a) Explain the working of servomotors in control system. (10)

- (b) Open loop transfer function $G(s)H(s) = \frac{K}{s(s+1)(s+2)}$. Using Rouths

Hurwitz criterion determine the value of K which causes sustained oscillation.

(10)

OR

- VI. (a) List the steps for constructing root locus. (10)

- (b) For the system with open loop transfer function

$$G(s) = \frac{K}{s(s+2)(s+4)}$$

Find -

- (i) centroid (ii) breakaway and break in points.

(10)

- VII. (a) Sketch the Bode plot for the unity feedback system whose open loop transfer function is

$$G(s) = \frac{50}{(s+1)(s+2)}$$

Comment on the stability of the system.

(15)

- (b) What are the frequency domain specifications. (5)

OR

- VIII. (a) State and explain Nyquist stability criterion. (10)

- (b) Explain M and N circles. (10)

- IX. (a) Derive the transfer function of a RC phase lead network. (10)

- (b) Explain PI, PD and PID control action. (10)

OR

- X. Design a phase lead compensation network for a system having open loop transfer function

$$G(s)H(s) = \frac{K}{s^2(1+0.05s)}$$

It is desired to have an acceleration error coefficient $Ka = 100 \text{ sec}^{-2}$ and suitable phase margin for stable operation.

(20)
